Experiment

ISPhO

A1-1

English

LED Color

Part A. Calibration (0.6 points)

A.1 (0.3pt)

Spectra saved: R.csv G.csv B.csv

A.2 (0.1pt)

Colour	N
Red	
Green	
Blue	

A.3 (0.2pt)

$\mathrm{slope} =$ $\mathrm{shift} =$

Part B. LED in air (5.3 points)

B.1 (0.1pt)

Range, V	R_V
600	
200	
20	
2	
200m	

B.2 (0.1pt)

$R_0 =$ $U_{100\,\mathrm{mA}} =$

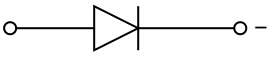
B.3 (0.3pt)

Circuit:

B.4 (0.6pt)

V_1 ( )	V_2 ( )

B.5 (0.1pt)

(1) +  –

(2) +  –

Correct polarity: (1) / (2)

B.6 (0.3pt)

Circuit:

B.7 (0.9pt)

V_1 ( )	V_2 ( )

B.8 (0.4pt)

Low-current mode:

$I =$

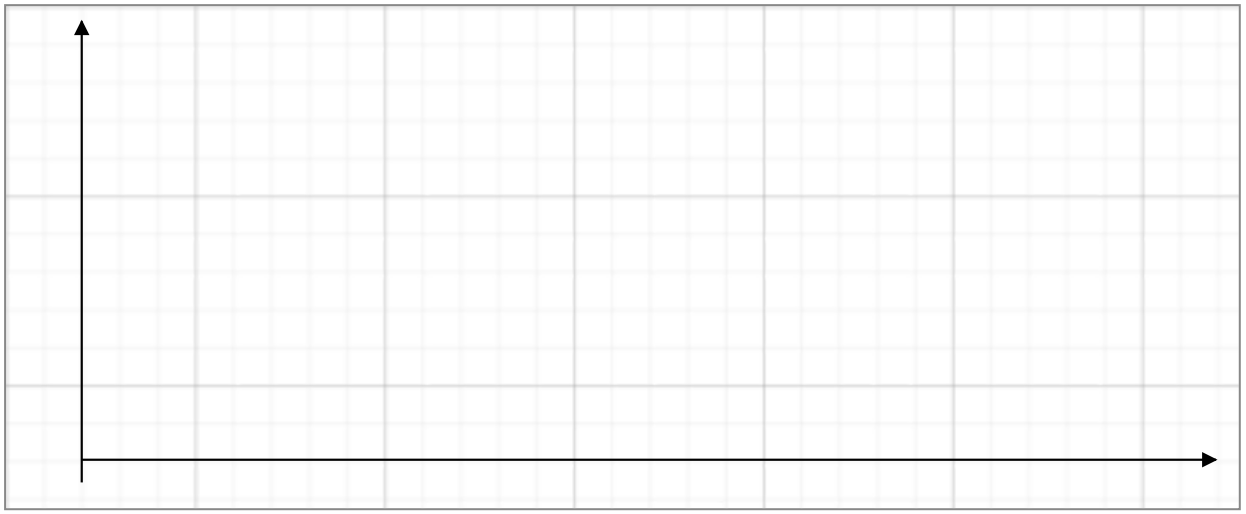
$U =$

High-current mode:

$I =$

$U =$

B.9 (0.2pt)
Plot:



B.10 (0.1pt)

$\eta_0 k_B T_0 / e =$

B.11 (0.3pt)

$\eta_{\text{room}} =$

B.12 (0.8pt)

I, mA	λ, nm

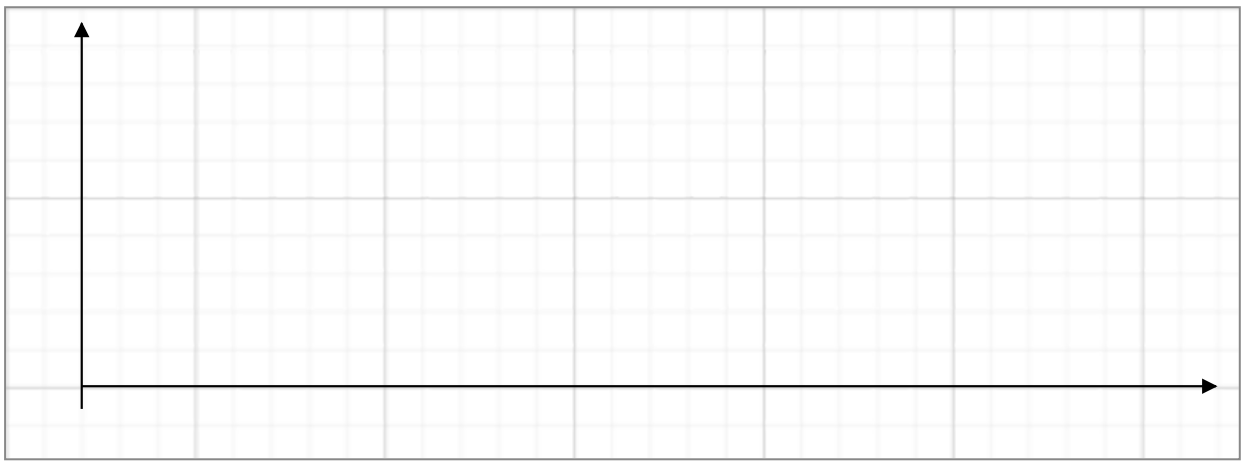
B.13 (0.8pt)

Time series saved:

B.14 (0.3pt)

t, s	λ, nm

Plot:



Part C. LED in water (3.5 points)

C.1 (0.9pt)

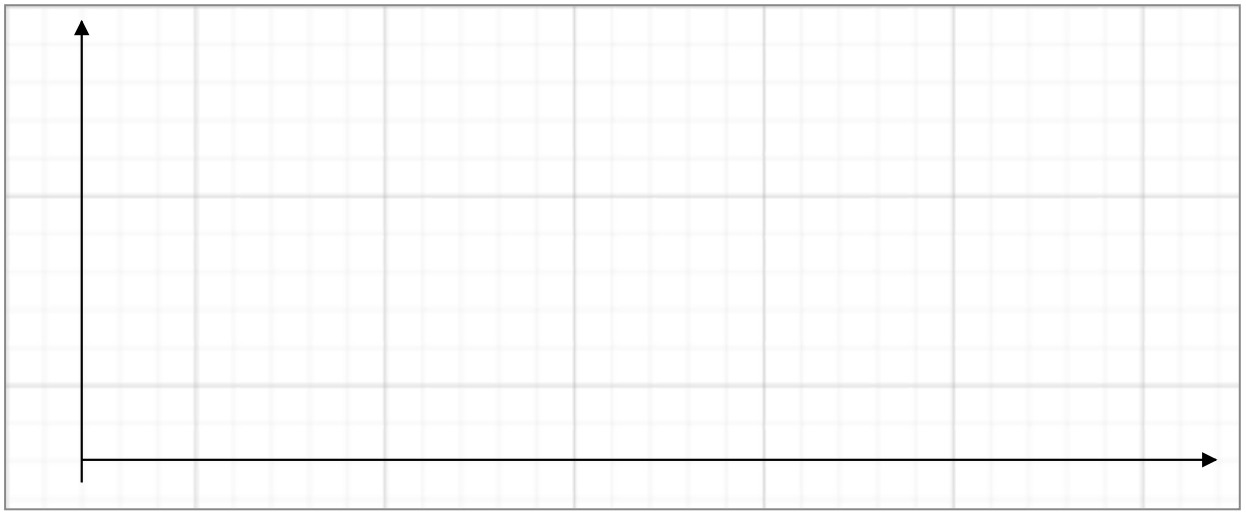
V_1 ( )	V_2 ( )

C.2 (0.2pt)

$I =$

$U =$

C.3 (0.5pt)
Plot:



C.4 (0.8pt)

I, mA	λ, nm

C.5 (0.8pt)

Time series saved:

C.6 (0.3pt)

t, s	λ, nm

Part D. LED in liquid nitrogen (6.1 points)

D.1 (0.1pt)

Spectrum saved:

D.2 (0.6pt)

V_1 ( )	V_2 ( )

D.3 (0.9pt)

V_1 ( )	V_2 ( )

D.4 (0.2pt)

Low-current mode:

$I =$

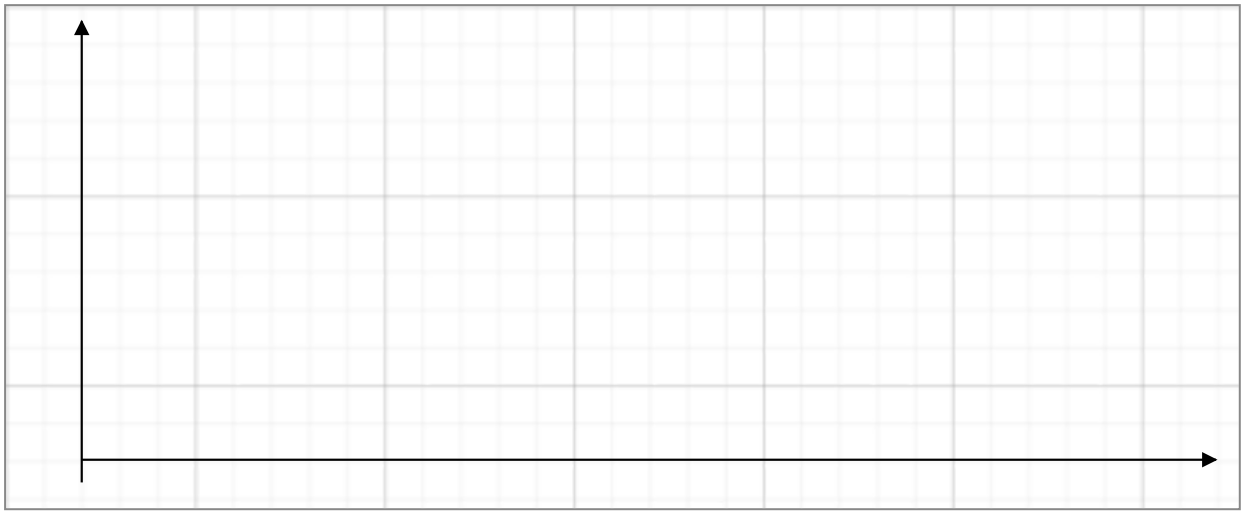
$U =$

High-current mode:

$I =$

$U =$

D.5 (0.2pt)
Plot:



D.6 (0.3pt)

$\eta_{\mathrm{N_2}} =$

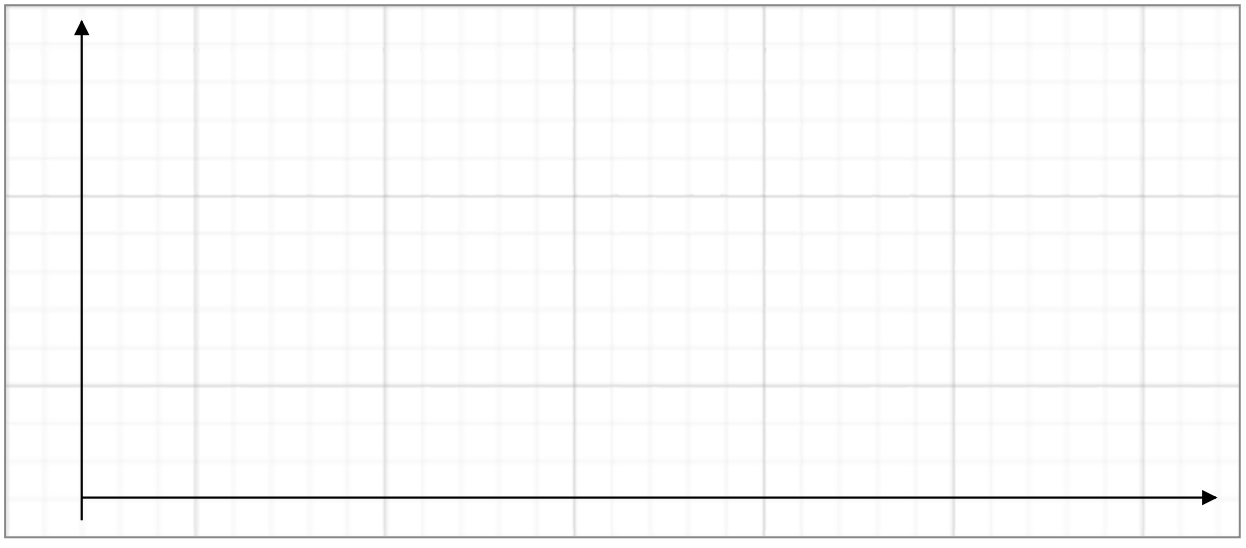
D.7 (1.0pt)

I, mA	Spectrum file

D.8 (0.5pt)

I, mA	λ, nm

D.9 (1.0pt)
Plot:



D.10 (0.6pt)

$\lambda_{N_2} =$ $\lambda_{\text{room}} =$

D.11 (0.7pt)

$E_0 =$ $\alpha =$

Part E. Calculation of calorimetric parameters (3.1 points)

E.1 (0.5pt)

(Formula) $\chi =$ (Value) $\chi =$

E.2 (0.5pt)

$\varkappa_{\text{air}}/\varkappa_{\text{water}} =$

E.3 (0.3pt)

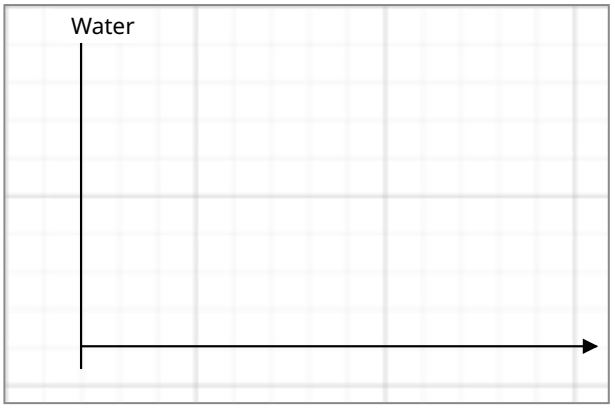
$T(t) =$

E.4 (1.0pt)

t, s (air)	T	t, s (water)	T

E.5 (0.5pt)
Plot:





E.6 (0.3pt)

$C_{\text{air}}/C_{\text{water}} =$

Part F. Temperature dependence of saturation current (1.4 points)

F.1 (0.6pt)

T	$\ln(I_0/\text{mA})$

Plot:



F.2 (0.6pt)

T	$\ln(I_0/\mathrm{mA})$

Plot:



F.3 (0.2pt)

Factor: